

Reviewer Pack — RSK Second-Row Length of f-Sorted Permutation Bounds Lifted ...

Entry a991258bde76 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Entry ID: a991258bde76 **Verdict:** INCONCLUSIVE **Recorded:** 2026-04-27 16:27:28 UTC

1. Conjecture

Field A (mathematical branch): Plactic monoid combinatorics via Robinson-Schensted-Knuth (RSK) insertion and Schensted’s theorem on Young tableau shapes (Lothaire, Stanley EC2 Ch. 7); the longest weakly-decreasing-subsequence statistic λ_2 of a permutation, accessed by patience sorting

Field B (complexity object): Deterministic decision-tree depth $Q^{dt}(f)$ of small Boolean functions $f: \{0,1\}^k \rightarrow \{0,1\}$, lifted to deterministic two-party communication complexity $D^{cc}(f \circ \text{IND}_b)$ via the Raz-McKenzie / Göös-Pitassi-Watson indexing-gadget lifting theorem

Statement:

For $f: \{0,1\}^k \rightarrow \{0,1\}$, let π_f be the permutation of $[0..2^k-1]$ obtained by stable-sorting the inputs by the key $(f(x), x)$, and let $\lambda(f) = (\lambda_1, \lambda_2)$ be the shape of the RSK insertion tableau $P(\pi_f)$. Conjecture: $\lceil \log_2(\lambda_2(f)+1) \rceil \leq Q^{dt}(f)$ for every Boolean f , with equality achievable by parity-like functions. By GPW lifting with indexing gadget IND_b , this entails $D^{cc}(f \circ \text{IND}_b) \geq \lceil \log_2(\lambda_2(f)+1) \rceil \cdot \log_2 b - O(\log k)$.

Rationale (proposer’s reasoning):

Schensted’s theorem makes $\lambda_2(\pi_f)$ the length of the longest weakly-decreasing subsequence of π_f , a global obstruction distinct from antichain width, matroid rank, or Möbius invariants — it captures how thoroughly the value-class boundary scrambles the lex order. Each decision-tree query can split the truth table into two halves but cannot shorten a long alternating descent chain by more than a factor of 2, suggesting Q^{dt} must grow logarithmically with λ_2 . The construction is a simple permutation statistic, side-steps the natural-proofs barrier (λ_2 is not a pseudorandom property — random f has $\lambda_2 \approx c \cdot 2^{\{k/2\}}$), is unaffected by algebrization (no oracle-relativizing structure), and RSK has never been deployed inside the GPW lifting framework.

Taxonomy category: LIFTING (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): a35a2da41675cbb3

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

Enumerate all 65536 Boolean functions f on $k=4$ variables. For each f , compute $Q^{dt}(f)$ via memoized DP and $\lambda_2(f)$ via RSK on $\pi_f = \text{stable-sort}([0..15], \text{key}=(f(x), x))$. Conjecture is SUPPORTED iff $\lceil \log_2(\lambda_2(f)+1) \rceil \leq Q^{dt}(f)$ for every one of the 65536 functions (zero violations); a single counterexample FALSIFIES.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.85	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.82	SAFE	SAFE
ALGEBRIZATION	SAFE	0.85	SAFE	SAFE
KARP_LIPTON	SAFE	0.93	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 8 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - RSK insertion tableau shape Boolean function decision tree complexity - longest decreasing subsequence permutation query complexity Boolean-Raz-McKenzie lifting indexing gadget Young tableau plactic communication complexity

Top relevant hits considered: - [<http://arxiv.org/abs/1808.06095v5>] Skew RSK and the switching on ballot tableau pairs - [<http://arxiv.org/abs/1810.08668v2>] Parity Decision Tree Complexity is Greater Than Granularity - [<http://arxiv.org/abs/2512.11833v1>] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [<http://arxiv.org/abs/math/0109079v2>] $GL(n, q)$ and Increasing Subsequences in Nonuniform Random Permutations - [<http://arxiv.org/abs/2309.01208v1>] Streaming and Query Once Space Complexity of Longest Increasing Subsequence - [<http://arxiv.org/abs/math/0506012v1>] Longest alternating subsequences of permutations - [<http://arxiv.org/abs/1904.13056v4>] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [<http://arxiv.org/abs/2405.02711v1>] The Role of AI in Peer Support for Young People: A Study of Preferences for Human- and AI-Generated Responses

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90s$ wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import sys
import random

def Q_dt(f):
    n = 16
    cache = {}

    def helper(x):
        if x not in cache:
            if len(x) == 1:
                return 0
            else:
```

```

        left = helper([f(xi, 0) for xi in range(n)])
        right = helper([f(xi, 1) for xi in range(n)])
        cache[x] = 1 + max(left, right)
    return cache[x]

return helper(tuple(range(n)))

def RSK_insertion_tableau(arr):
    shape = [0]
    stack = []
    for x in arr:
        i = bisect_right(stack, x)
        if i == len(stack):
            stack.append(x)
            shape[-1] += 1
        else:
            stack[i] = x
    return shape

def run_trial(seed: int) -> dict:
    random.seed(seed)

    k = 4
    n = 2**k

    def f(x, y):
        return (x >> (k - y)) & 1

    metric_name = "slack"
    instances_tested = 0
    total_slack = 0.0
    conjecture_holds = True
    counterexample = ""

    for i in range(n):
        f_i = [f(i, j) for j in range(k)]
        pi_f = sorted(range(n), key=lambda x: (f(x, j) for j in range(k)))
        lambda_2_f = RSK_insertion_tableau(pi_f)[-1]

        slack = max(0, Q_dt(f_i) - math.ceil(math.log2(lambda_2_f + 1)))
        total_slack += slack
        instances_tested += 1

        if slack > 0:
            conjecture_holds = False
            counterexample = f"Function {i} has a slack of {slack}"

    mean_slack = total_slack / instances_tested

    return {
        "metric_name": metric_name,
        "metric_value": mean_slack,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }
}

```

```

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [11, 23, 37, 53, 71]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_slack = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_slack} std=0.0 support_fraction=1.0")
    elif any(not r["conjecture_holds"] for r in results) and support_fraction >= 0.8:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE insufficient data")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4eae8e14.py", line 89, in <module>
    result = run_trial(seed)
              ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4eae8e14.py", line 63, in run_trial
    pi_f = sorted(range(n), key=lambda x: (f(x, j) for j in range(k)))
              ~~~~~^~~~~~
TypeError: '<' not supported between instances of 'generator' and 'generator'

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed with a `TypeError` before producing any data — the sort key was incorrectly written as a generator expression rather than a tuple, so no functions were evaluated and no RESULT line was produced. | next: Fix the key in the sort to `key=lambda x: (f(x), x)` (using the precomputed truth table `f[x]`) and rerun the full enumeration over all 65536 functions on `k=4`.

11. Audit log (LLM calls)

Total LLM calls: 8

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	claude_max	opus	0	0	21110
2	propose	claude_max	opus	0	0	198126
3	preregistration	claude_max	opus	0	0	8404
4	novelty	claude_max	opus	0	0	3843
5	novelty	claude_max	opus	0	0	8577
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11670
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8631
8	judge	claude_max	opus	0	0	4647

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 265007 ms total latency. Provider mix: {'claude_max': 6, 'ollama_remote': 2}

(full prompt+response transcripts available in research/audit/a991258bde76.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/a991258bde76.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/a991258bde76.tar.gz (if generated)